Rotate Image (medium)

Problem Statement

Given an `n x n` 2D matrix, modify a square matrix by `rotating` it `90 degrees` in a `clockwise` direction.

Note: This rotation should be done `in-place`, meaning the transformation must occur within the original matrix without using any additional space for another matrix.

Examples

- **Example 1**:
  - **Input**: matrix =

```
[[1,2],
 [3,4]]
```

- **Expected Output**:

```
[[3,1],
 [4,2]]
```

- **Justification**: After rotating the 2x2 matrix 90 degrees to the right, the element at the top left (1) moves to the top right, the top right (2) moves to the bottom right, the bottom right (4) moves to the bottom left, and the bottom left (3) moves to the top left.

- **Example 2**:

  - **Input**: matrix =

```
[[5,1,9],
 [2,4,8],
 [13,3,6]]
```

- **Expected Output**:

```
[[13,2,5],
 [3,4,1],
 [6,8,9]]
```

- **Justification**: Rotating the 3x3 matrix 90 degrees to the right repositions the first row to the last column, the second row to the middle column, and the third row to the first column.

- **Example 3**:

  - **Input**: matrix =

```
[[10,11,12,13],
 [14,15,16,17],
 [18,19,20,21],
 [22,23,24,25]]
```

- **Expected Output**:

```
[[22,18,14,10],
 [23,19,15,11],
 [24,20,16,12],
 [25,21,17,13]]
```

- **Justification**: The matrix is rotated by 90 degrees in the clockwise direction.

Try it yourself

Try solving this question here:

Java

```java
import java.util.Arrays;

public class Solution {
    public int[][] rotate(int[][] matrix) {
        // TODo: Write Your COde Here.
        return matrix;
    }
}
```

Test Case | Run Code Result | Stdout

[[1,2,3],[4,5,6],[7,8,9]]

Use Example Testcases | Run | Submit

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Solution: Rotate Image

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